



CIE ACTIVITY 1

“DESIGN & SIMULATION OF VARIOUS VITAL PERIPHERALS INTERFACING WITH 8051 AND PIC18F4550 MICROCONTROLLER USING PROTEUS SOFTWARE TOOLS”.

Activity Description:

Students will **Design** and **simulate** any one of the embedded-system problem statements given below by developing the corresponding system design and implementing the derived embedded application using either an **8051** or **PIC18F4550** microcontroller. **Develop** the required interfacing circuit, simulate the designed system using Proteus, and **analyse** the implementation based on the selection of peripherals, microcontroller selection criteria, and the obtained simulation results.

Problem Statements:

1. Industrial Motor Control System
2. Home Automation Using suitable microcontroller
3. Industrial Fault Monitoring System
4. Weather Monitoring System
5. Smart Traffic Light Control
6. Smart Door Locking System
7. Smart Farm Monitoring and Irrigation System
8. Industrial Voltage and Current Monitoring
9. LPG Gas Leakage Detection and Alert System
10. Patient Health Monitoring System
11. Interface 7 segment display to count 00 to 99
12. Generate Square wave, Triangular, Ramp, Sine wave using DAC.
13. Interface Stepper motor and rotate motor clockwise and anticlockwise wise in 90
14. Interface Stepper motor and rotate motor clockwise and anticlockwise wise in 360.
